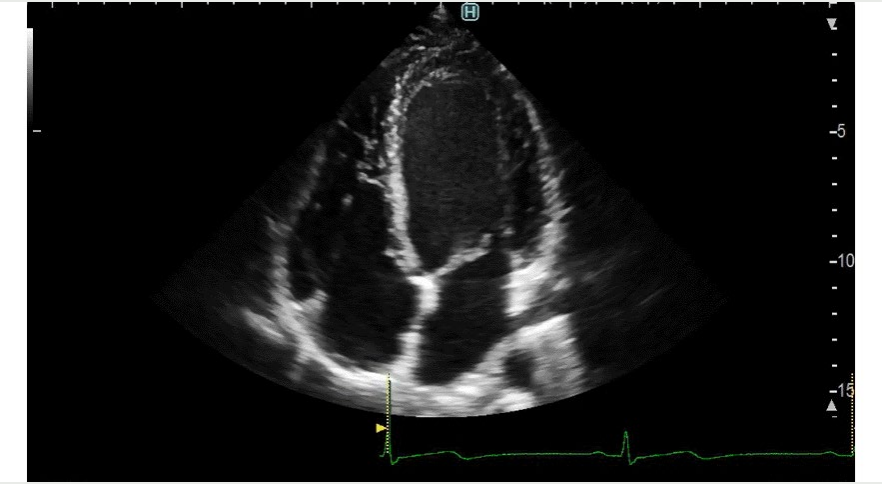
Dehazing Echocardiography for Difficult-to-Image Patients: ROI-Aware GAN Pipeline

Pushpendra Singh
King's College London
pushpendra.1.singh@kcl.ac.uk

Navchetan Awasthi
University of Amsterdam
n.awasthi@uva.nl



MICCAI2025
Daejeon
REPUBLIC OF KOREA

Motivation

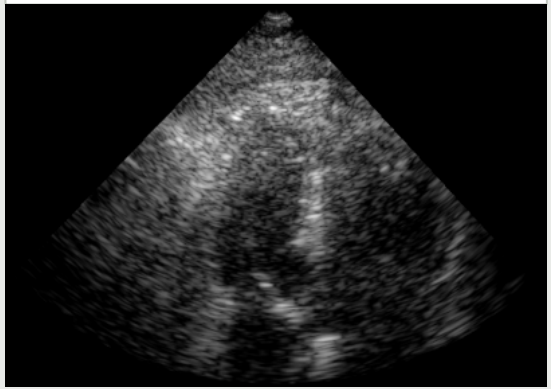
Echocardiography: real-time, cost-effective cardiac imaging.

Problem: up to 20% of patients → poor quality due to acoustic clutter/ haze.

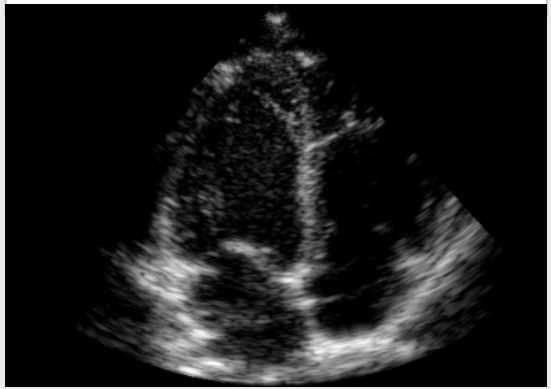
Consequence: nondiagnostic exams → fallback to costly MRI/ CT.

Need: robust dehazing to improve clarity, usability, and AI reliability.

*Echocardiograms of difficult -to-image patient

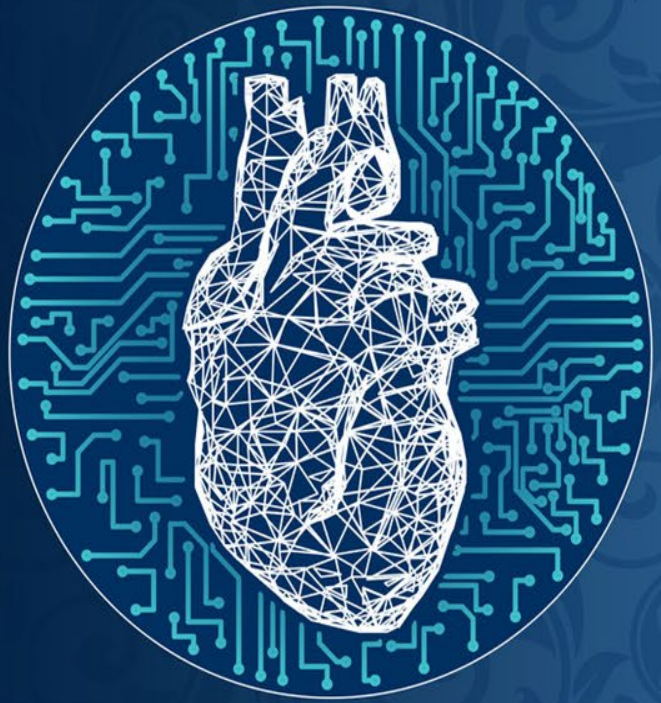


*Echocardiograms of easier-to-image patient



* Stevens, T. S. W., Meral, F. C., Yu, J., Apostolakis, I. Z., Robert, J.-L., & van Sloun, R. J. G. (2024). Dehazing Ultrasound using Diffusion Models. IEEE Transactions on Medical Imaging, 43(10), 3546–3558. Article 10423849. https://doi.org/10.1109/TMI.2024.3363460

Challenge Context

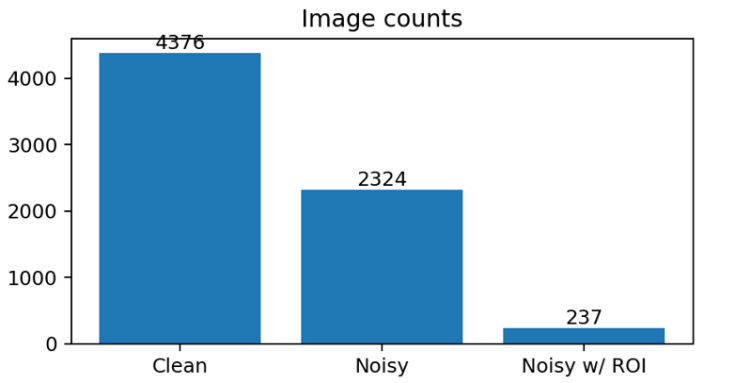


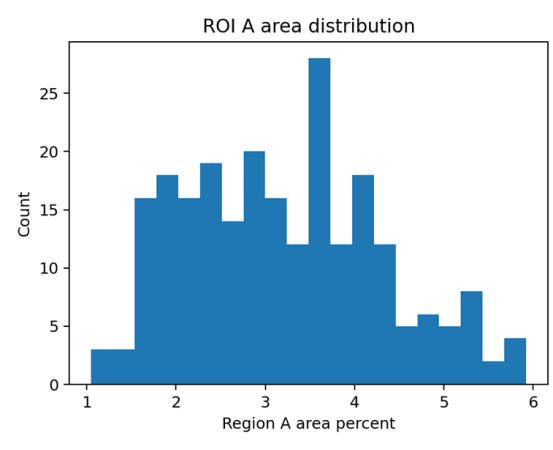
DehazingEcho2025

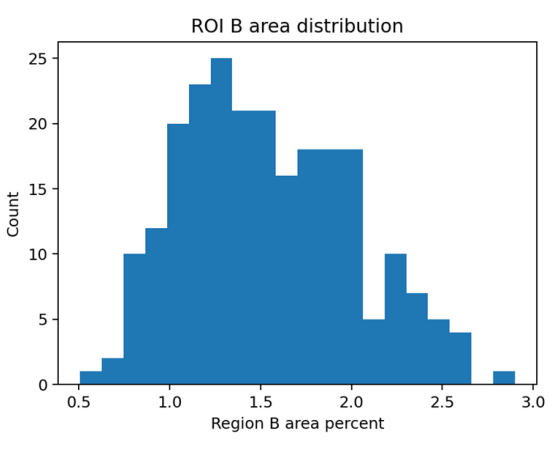
Dehazing Echocardiography Challenge:

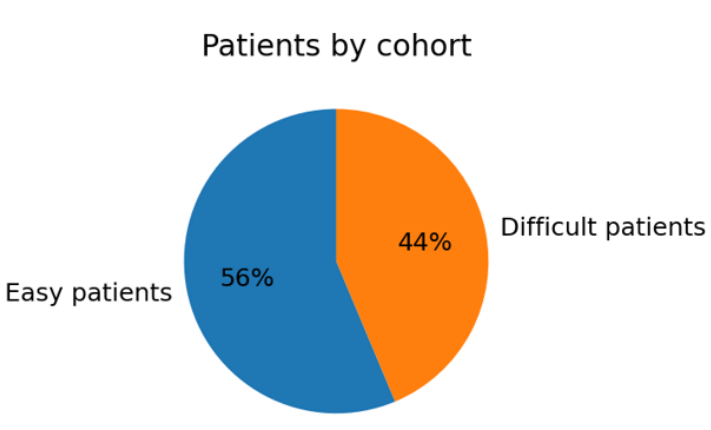
Enhancing echocardiographic image quality for difficult-to-image patients

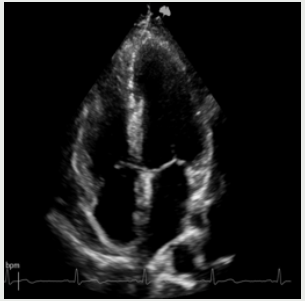
Challenge Context

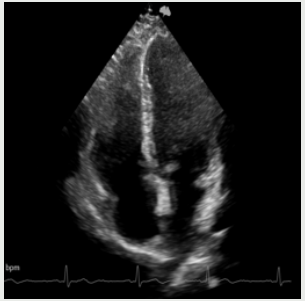


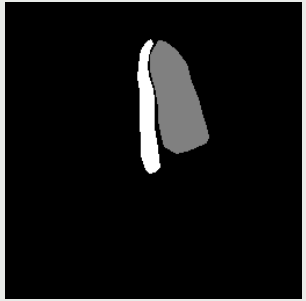






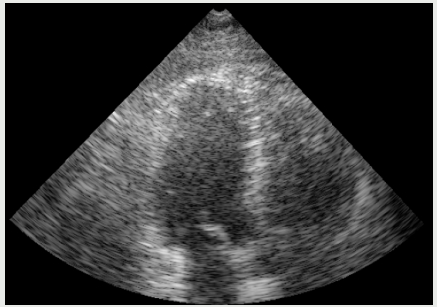




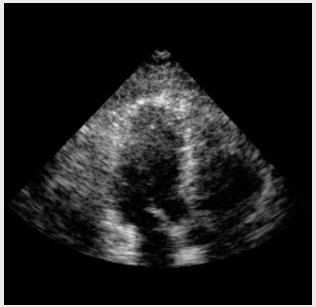


Related Work

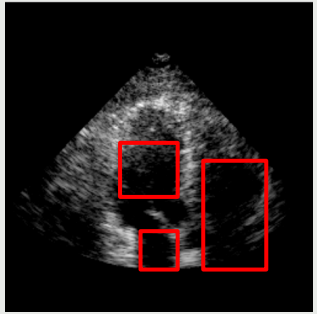
Diffusion Model



a) Noisy Echo [1]

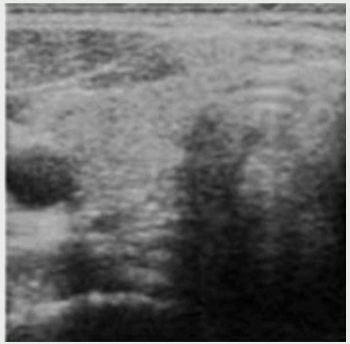


b) Dehazing using diffusion [1]

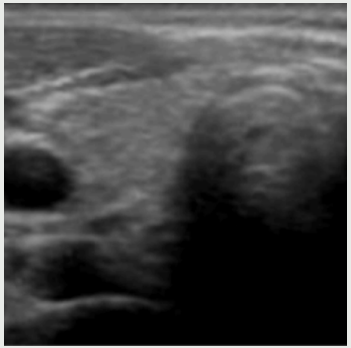


c) Further Dehazing using our method

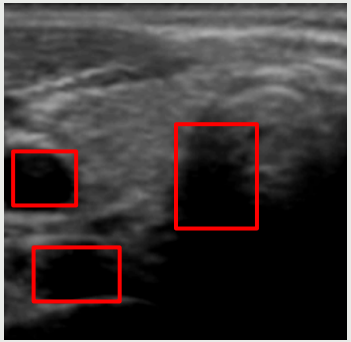
CycleGAN



a) Noisy Echo [2]



b) Dehazing using CycleGAN [2]



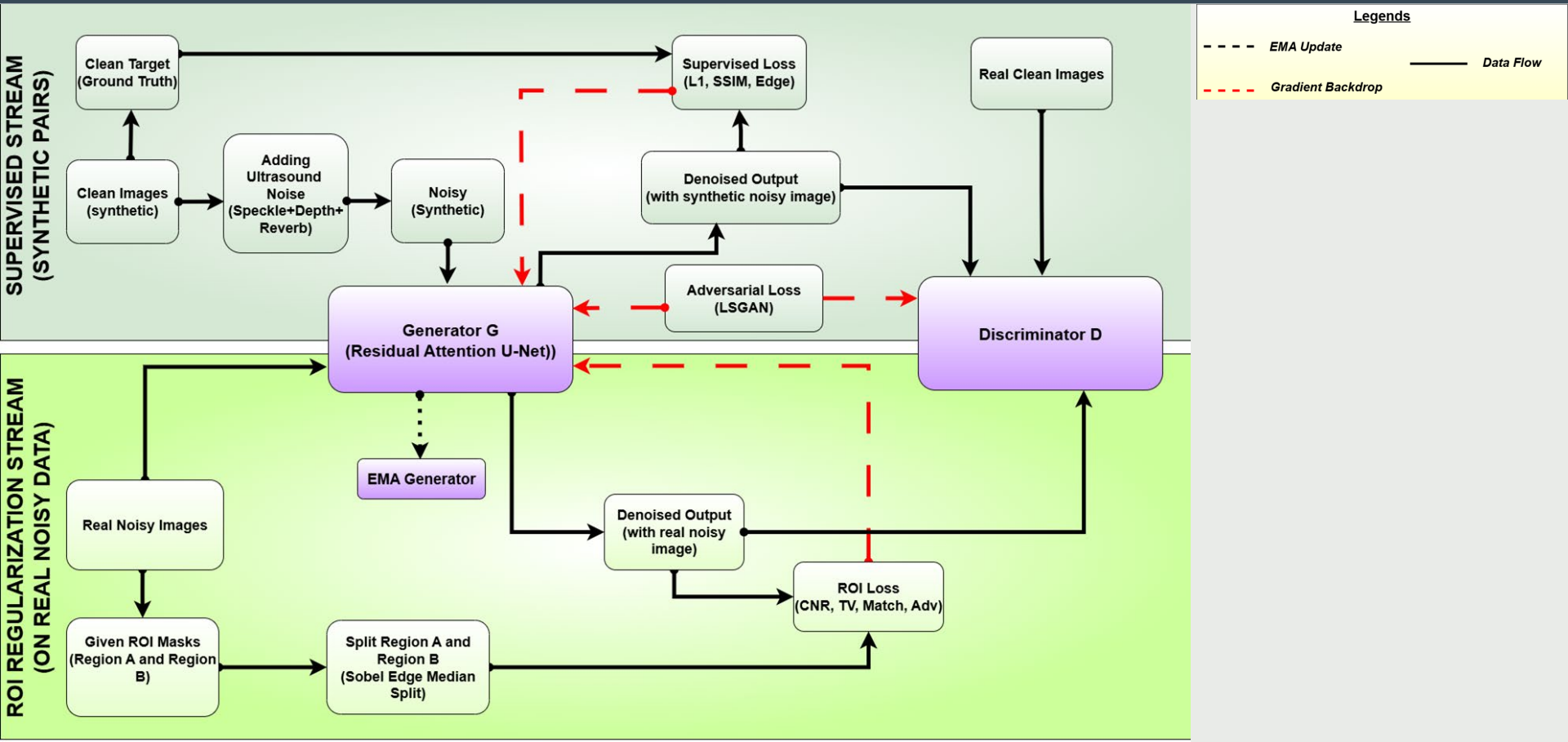
c) Further Dehazing using our method

[1] Stevens, T. S. W., Meral, F. C., Yu, J., Apostolakis, I. Z., Robert, J.-L., & van Sloun, R. J. G. (2024). Dehazing Ultrasound using Diffusion Models. IEEE Transactions on Medical Imaging, 43(10), 3546-3558. Article 10423849. https://doi.org/10.1109/TMI.2024.3363460

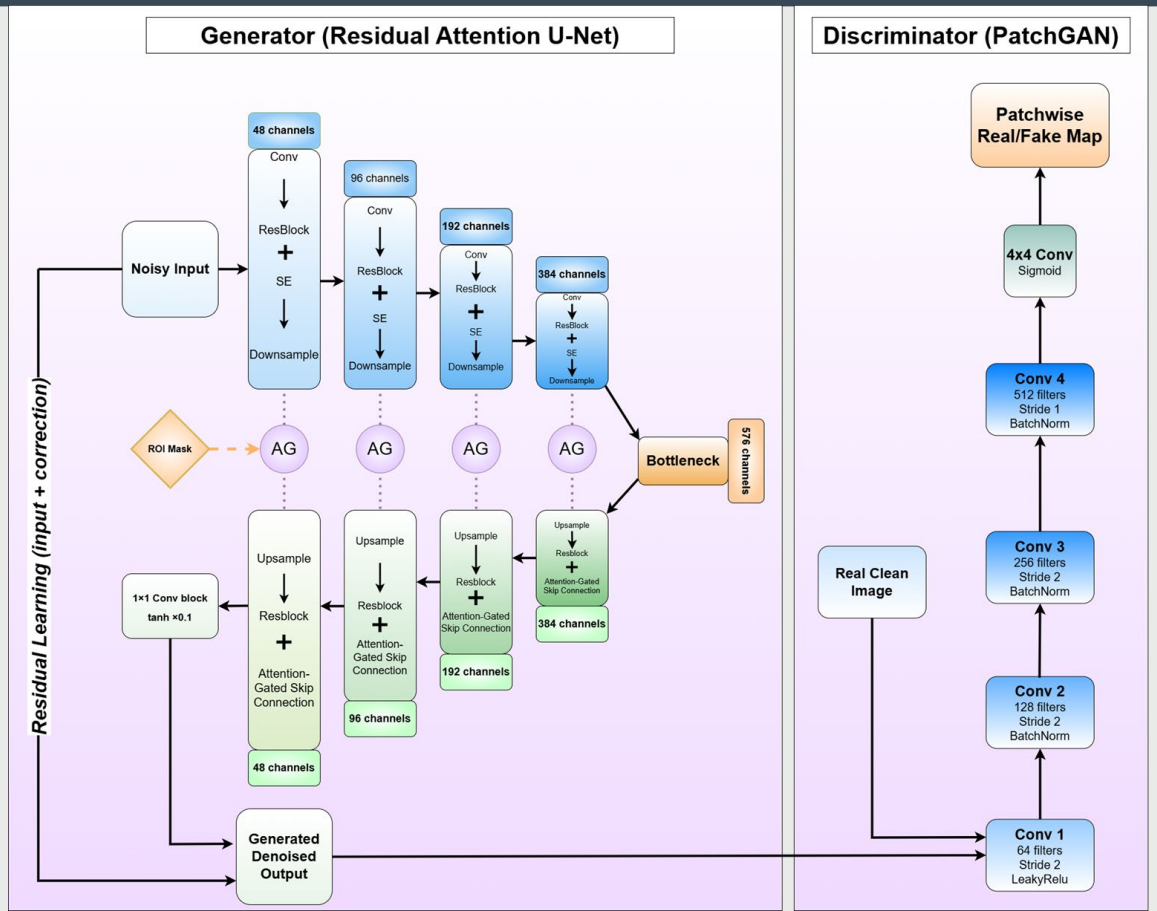
[2] Huang, L., Zhou, Z., Guo, Y., & Wang, Y. (2022). A stability-enhanced CycleGAN for effective domain transformation of unpaired ultrasound images. Biomedical Signal Processing and Control, 77, 103831 https://doi.org/10.1016/j.bspc.2022.103831

Pipeline Overview

Two-Stream ROI-Aware GAN Pipeline



Network Architecture



Loss Functions

supervised step

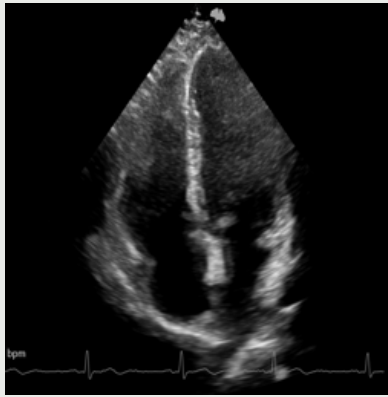
$$\mathcal{L}_G^{\text{sup}} = \underbrace{W_{\text{ADV}}\mathcal{L}_{\text{adv}}}_{0.02} + \underbrace{W_{L1}\mathcal{L}_{L1}}_{0.6} + \underbrace{W_{\text{SSIM}}\mathcal{L}_{\text{SSIM}}}_{1.0} + \underbrace{W_{\text{EDGE}}\mathcal{L}_{\text{edge}}}_{0.2}$$

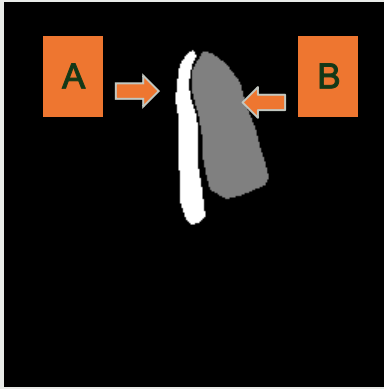
roi step

$$\mathcal{L}_G^{\text{roi}} = \underbrace{W_{\text{ROI_ADV}}\mathcal{L}_{\text{adv}}}_{0.02} + \underbrace{W_{\text{ROI_CNR}}\mathcal{L}_{\text{CNR}}}_{0.4} + \underbrace{W_{\text{ROI_TV}}\mathcal{L}_{\text{TV}}^{B}}_{10^{-4}} + \underbrace{W_{\text{ROI_MATCH_A}}\mathcal{L}_{\text{match}}^{A}}_{0.1}$$

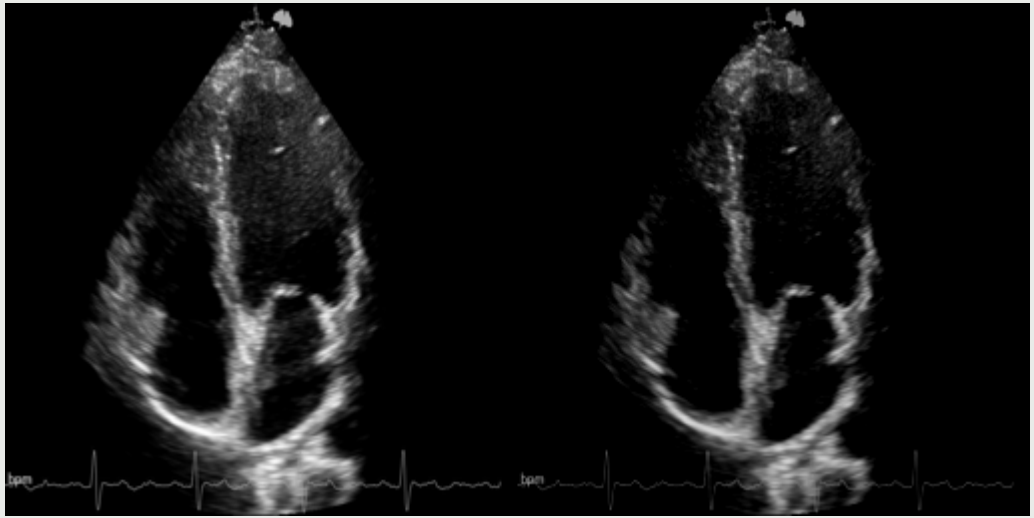
discriminator step

$$\mathcal{L}_D = \frac{1}{2}\|D(x_{\text{real}}) - 1\|_2^2 + \frac{1}{2}\|D(x_{\text{fake}})\|_2^2 + \mathcal{R}_{\text{R1}}$$





Method Impact: One - Epoch Training



Quantitative Results

Metrics	
FID	53.78
CNR (A vs B)	1.021
gCNR	0.571
KS (Region A)	0.253
KS (Region B)	0.416
LV Segmentation Dice	0.9064
LV Segmentation ASD (px)	1.7

Discussion

Strengths	Limitations
• ROI-guided • 1 epoch convergence • Good generalization	• Scores still improvable • Requires ROI masks

Conclusion

- ROI-aware GAN → effective dehazing in 1 epoch.
- Strong preservation of cardiac structures.
- Ready for real-world adaptation.

Thank You!

Feel free to drop a message to discuss the method and future improvements in detail.

Pushpendra: pushpendra.1.singh@kcl.ac.uk

Dr. Navchetan Awasthi: n.awasthi@uva.nl